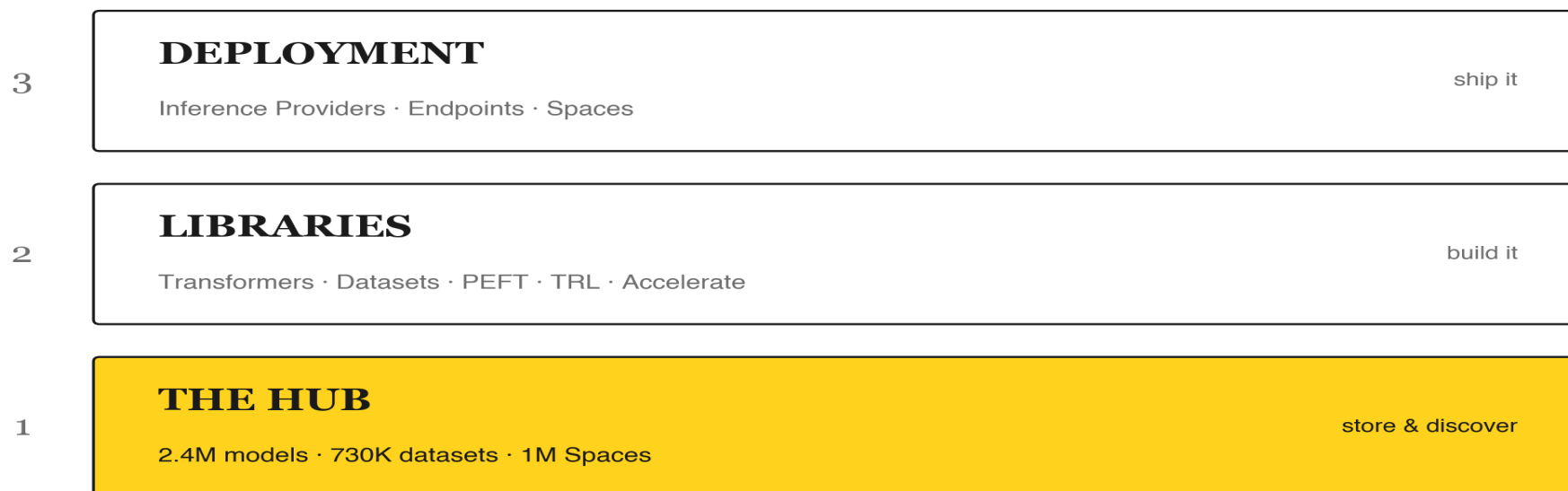

🤗 A LECTURE

Hugging Face

The mental model.



+ a long tail of extras — smolagents · LeRobot · Trackio · OpenEnv

Three layers — store, build, ship — plus a long tail of extras.

Scale (Jan 2026).

2.4M+

models

730K+

datasets

~1M

Spaces

Layer 1

The Hub.

Models · datasets · Spaces — git for weights.

Model & dataset cards.

- License — gated? commercial?
- Intended use, and limitations.
- How to evaluate before you download.

huggingface_hub — the plumbing.

```
hf_hub_download(...)
push_to_hub(...)
login()

snapshot_download(...)
```

The plumbing every other library sits on.

Safetensors — the security beat.

- Replaced pickle.
- Pickle = arbitrary code execution.
- `.safetensors` = safe, fast, zero-copy.

Xet — the new storage backend.

- Chunk-level deduplication.
- Replacing Git-LFS.
- Faster pulls on large, slowly-changing repos.

Layer 2

Core libraries.

The build stack.

Transformers.

- Load any model in about three lines.
- `pipeline()` · `AutoModel` · `AutoTokenizer`.
- Text · vision · audio · multimodal.

Transformers v5 (Dec 2025).

- First major release in five years.
- PyTorch-first — TF and Flax dropped.
- Unified AttentionInterface.
- Quantization is first-class.

Datasets.

- Stream TB-scale without downloading.
- Parquet + memory-mapping.
- `load_dataset(..., streaming=True)`.

Tokenizers.

- Rust backend.
- BPE · WordPiece · Unigram.
- Class 3 — the real thing.

Accelerate.

- Same code: one GPU → many.
- FSDP · DeepSpeed · fp8.
- Device-agnostic.

PEFT.

- LoRA / QLoRA.
- An adapter is a few MB, not GB.
- Classes 9a / 9b / 10a.

TRL.

— SFTTrainer · DP0Trainer · GRP0Trainer.

— Classes 9b / 10a / 10b.

— v1 shipped March 2026.

Diffusers.

- Image · video · audio generation.
- The branch we haven't touched.
- Same Hub, same patterns.

Optimum · Sentence-Transformers · timm.

Optimum

ONNX · TPU · Trainium

Sentence-Transformers

embeddings (RAG)

timm

vision backbones

Layer 3

Deployment.

Three confusingly-named products.

The three surfaces.

01 Serverless (hf-inference)

02 Inference Providers

03 Inference Endpoints

Serverless.

- Free, rate-limited — a few hundred / hour.
- Prototyping only.
- PRO (\$9/mo) raises the limits.

Inference Providers.

- An OpenAI-compatible gateway.
- Routes to Groq · Together · Fireworks · Cerebras.
- InferenceClient – a two-line swap from OpenAI.

Inference Endpoints.

- A dedicated GPU per model.
- ~\$0.50/hr+, scale-to-zero.
- vLLM default · SGLang for RAG · TGI in maintenance.

Spaces + Gradio.

- Ship a model as a web app.
- Gradio · Streamlit · static.
- ZeroGPU — 25 min/day on an H200 with PRO.

Layer 4

The extras.

What's new — and where it's going.

smolagents.

- Agents in a few lines.
- A code-first agent loop.
- vs. LangGraph (Class 15).

AutoTrain · Liteval · Evaluate.

AutoTrain

low-code fine-tuning

Liteval / Evaluate

the eval harness

Leaderboards

Class 11

Trackio.

— wandb drop-in: `import trackio as wandb.`

— Local-first, free.

— Dashboard on a Space.

OpenEnv.

- Agentic RL execution environments.
- Class 11 — RL envs.
- Step / reset / close, Docker-isolated.

LeRobot.

- Real-world robotics in PyTorch.
- Imitation + RL.
- SmolVLA foundation model.

Transformers.js + bitsandbytes.

Transformers.js

inference in the browser

bitsandbytes

quantization